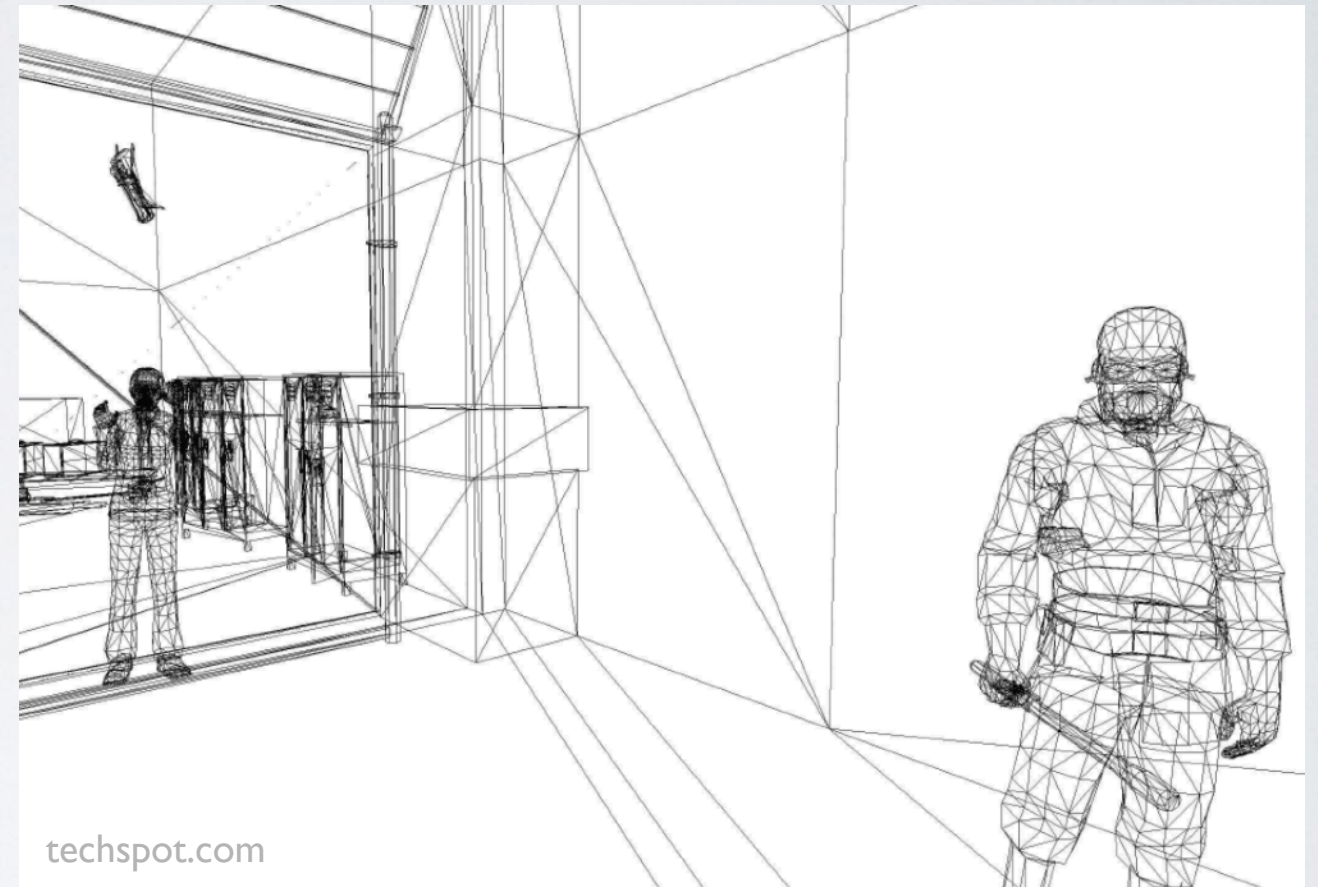
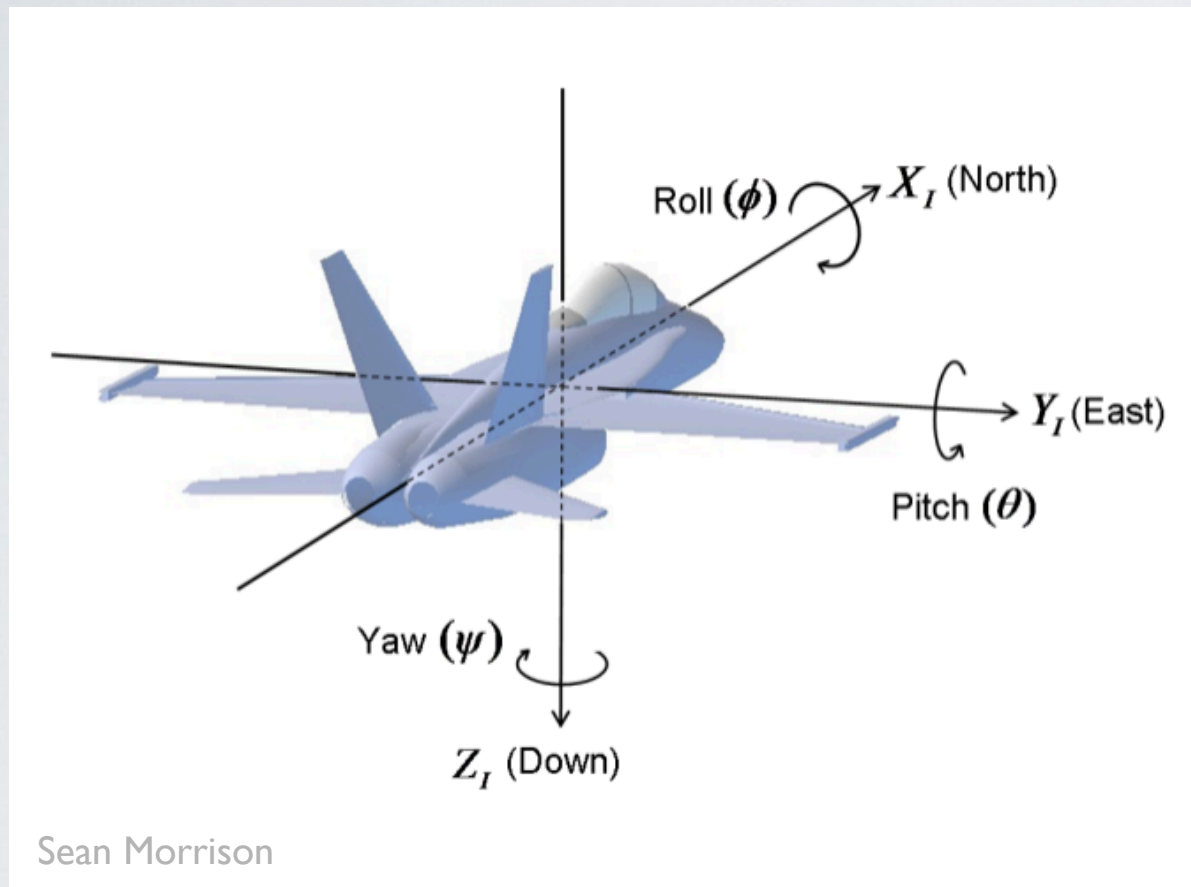


Week 5 - #1

Linear Algebra (V), Tensor Analysis (I)



Today: Ch 3, 10.1-10.4

Next Class: Ch 4

Ji-hoon Kim (Seoul National University)

Rudimentary Mathematical Methods of Physics (Fall 2026): Quiz #6

— [open book and open note, “and” cellphone or laptop, drop it off as you leave the class] —

Please write down your name and student ID in the top right corner. (0.0 pt: no paper found with your name / 0.5 pt: paper found with your name and some answers / 1.0 pt: good answers)

1. Boas, Chapter 3, Section 11, Problem 13
2. Study and briefly explain how matrix operations are used in 3-dimensional rendering in various fields such as game design, immersive visualization, augmented reality, and others.



Rudimentary Mathematical Methods of Physics (Fall 2026): Suggested Problems in Chapter 4, Boas, 3rd ed.

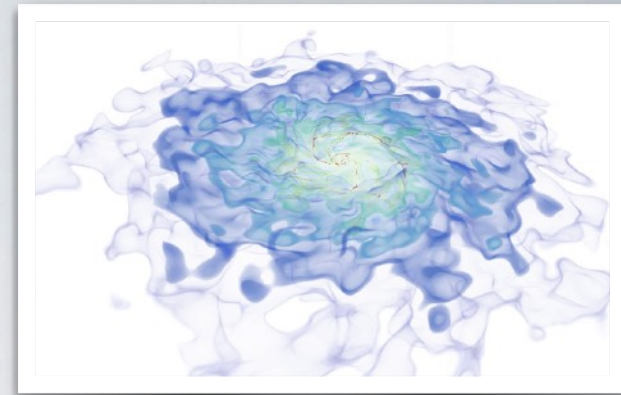
The problems I suggest you to take a deeper look into include, but are not limited to, the following.
The class homework assignments will mainly be from this list.

- Section 01: Problems 7, 8, 9, 15, 16
- Section 02: Problems 1, 2, 3
- Section 04: Problems 2, 3, 7, 15
- Section 05: Problems 1, 2, 6, 8
- Section 06: Problems 1, 4, 9
- Section 07: Problems 2, 4, 7, 10, 16, 25, 28
- Section 08: Problems 2, 8, 11, 13
- Section 09: Problems 2, 5, 9, 11
- Section 10: Problems 2, 5, 10, 14
- Section 11: Problems 1, 4, 6, 7, 8
- Section 12: Problems 2, 5, 10, 13, 16, 19
- Section 13: Problems 6, 9, 16, 27, 29

Rudimentary Mathematical Methods of Physics (Fall 2026): Suggested Problems in Chapter 5, Boas, 3rd ed.

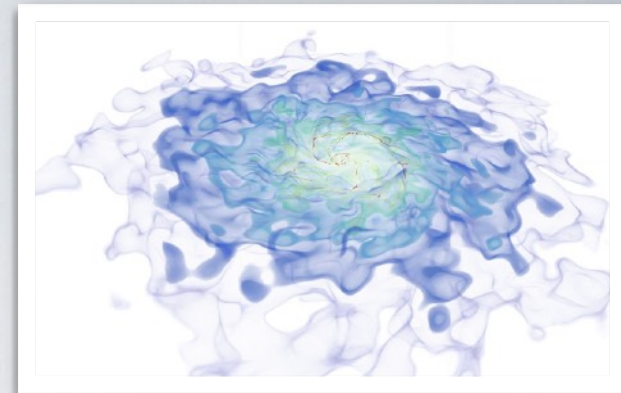
The problems I suggest you to take a deeper look into include, but are not limited to, the following.
The class homework assignments will mainly be from this list.

- Section 01: Problems 2, 4, 5
- Section 02: Problems 3, 5, 9, 11, 13, 16, 20, 22, 26, 39, 41
- Section 03: Problems 7, 12, 13, 17, 18, 19, 20, 21, 25
- Section 04: Problems 2, 4, 7, 14, 16, 19, 20
- Section 05: Problems 1, 5
- Section 06: Problems 4, 6, 11, 18, 23, 25, 27



HW #2 has been posted!

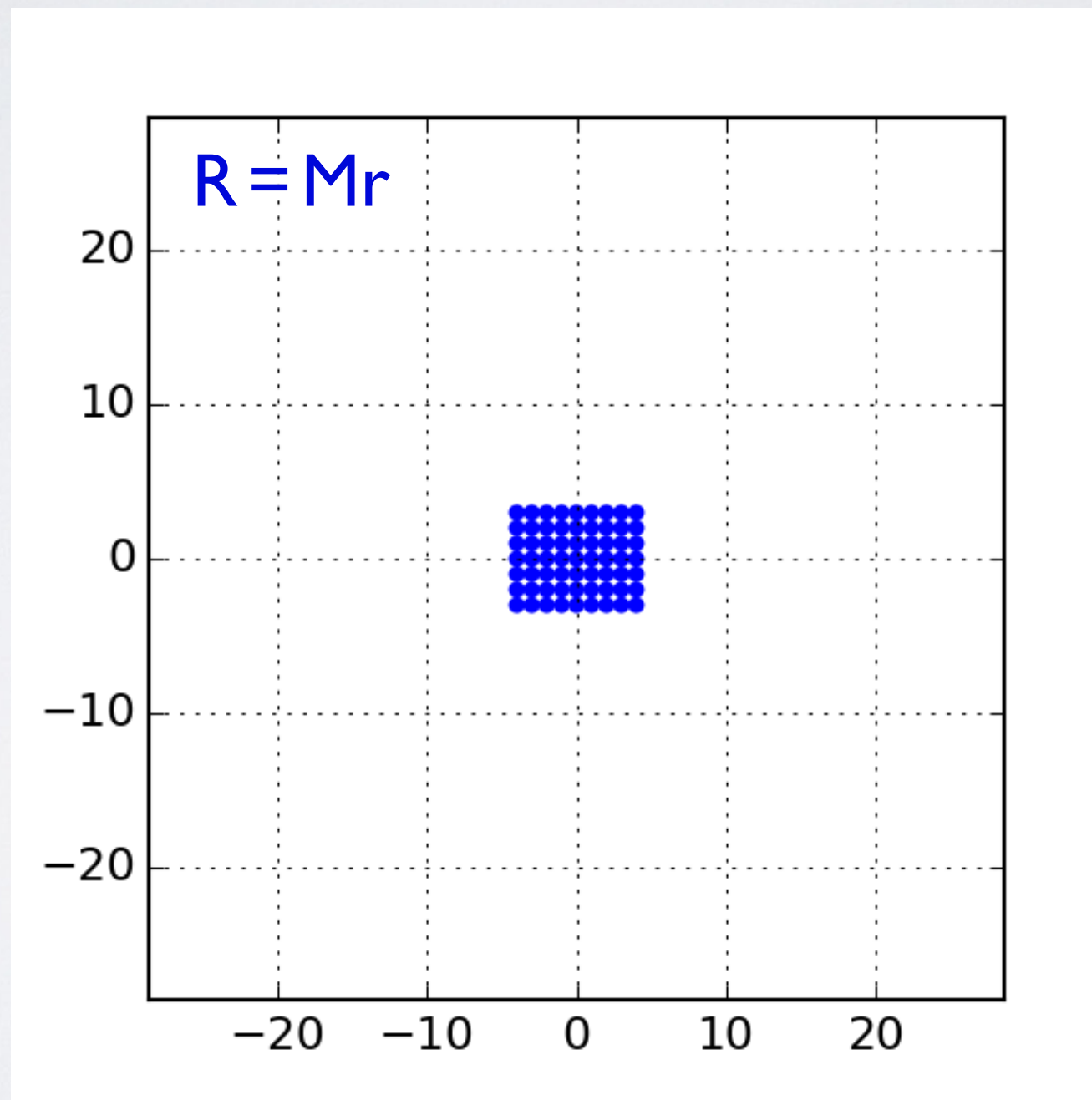
(Posted on jihoonkim.org, Due: **Oct. 2 (Fri), 23:00pm**,
Grader TA this time: 신찬미, chanmi0412@snu.ac.kr)



Meaning of Diagonalization

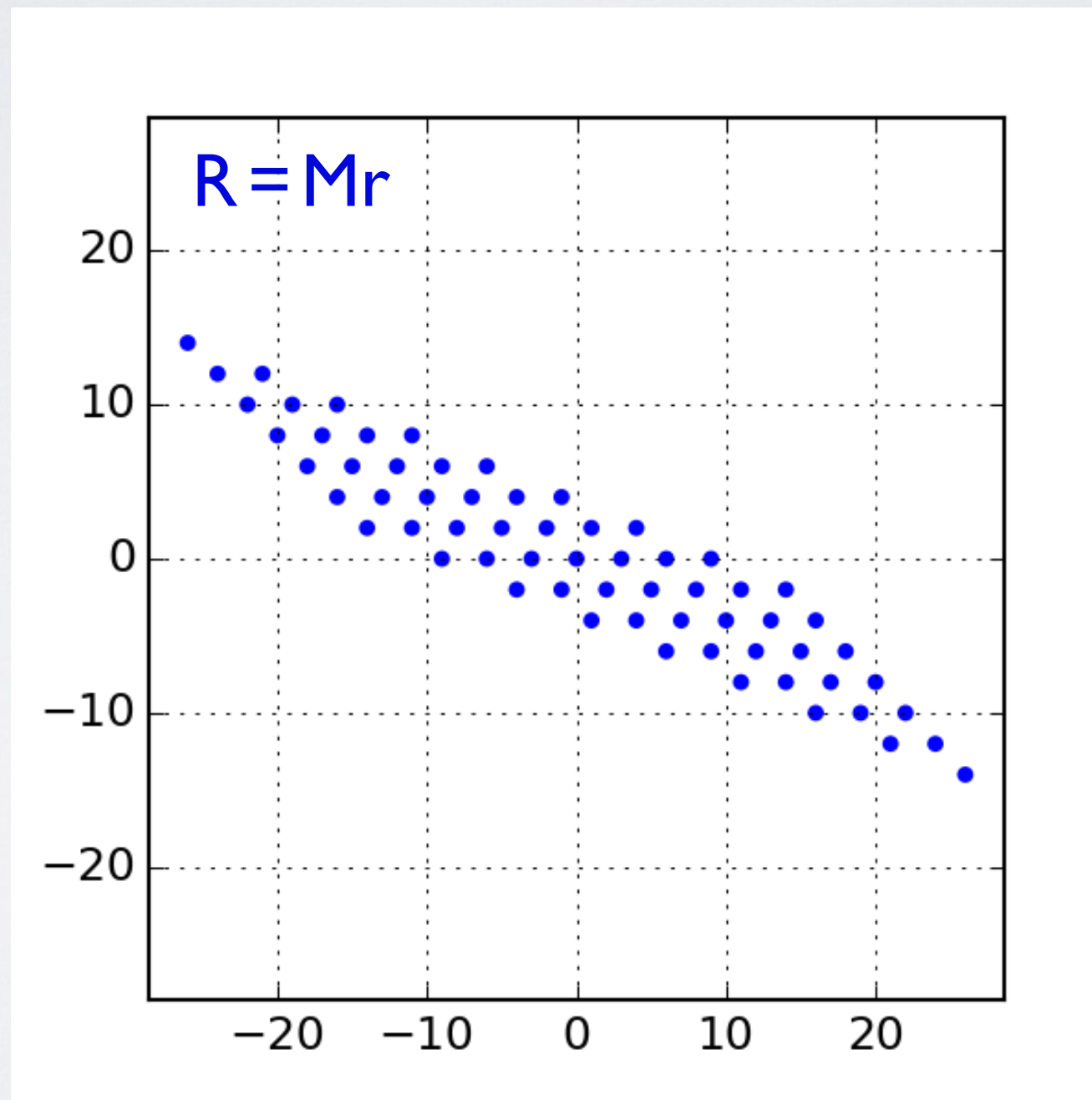
Meaning of Diagonalization (Ch 3.1 I)

- Matrix C^{-1} **changes** the old set of coordinate axes (x, y) to a new, convenient set of coordinate axes (x', y') .



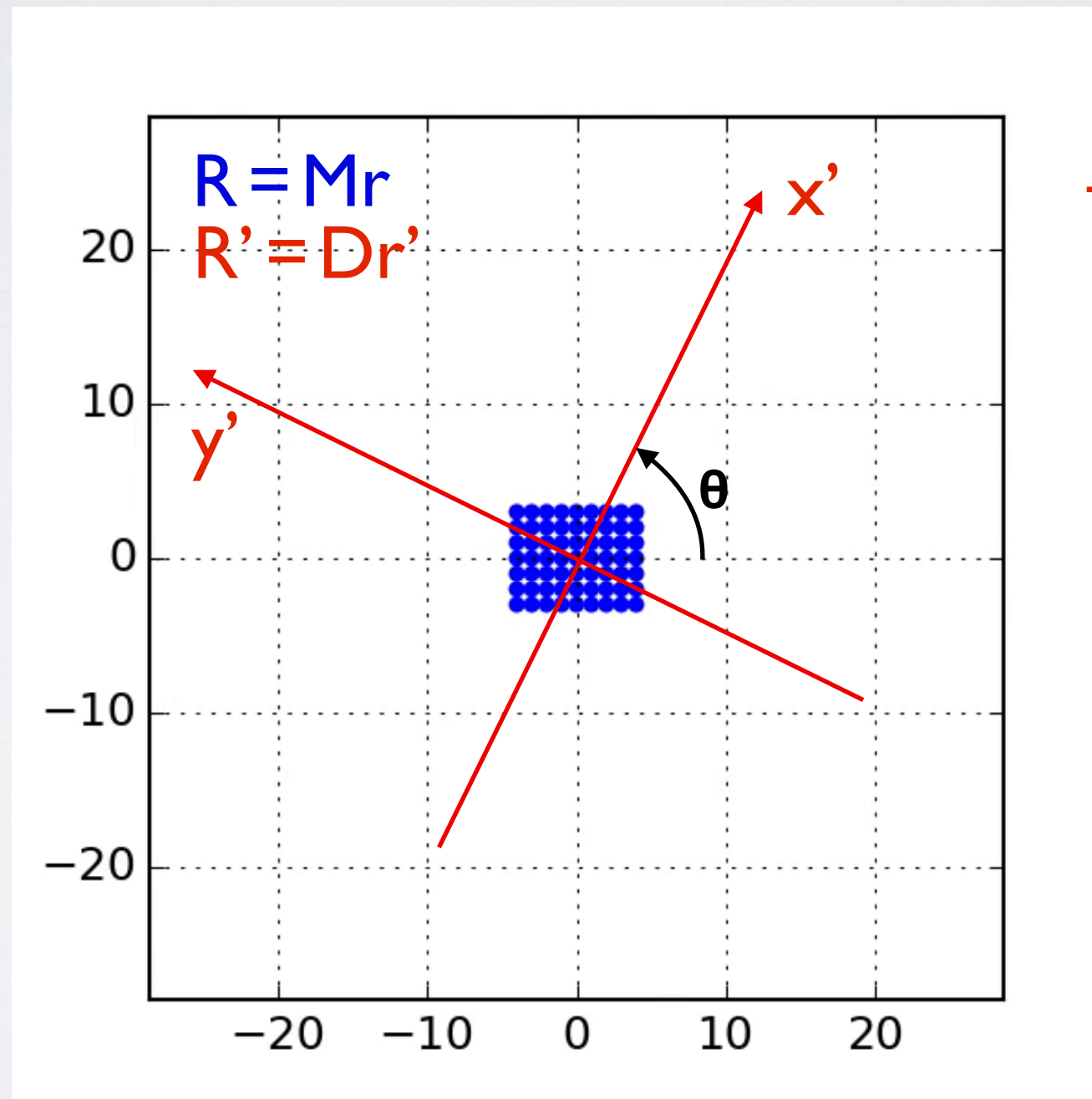
Meaning of Diagonalization (Ch 3.1 I)

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Meaning of Diagonalization (Ch 3.1 I)

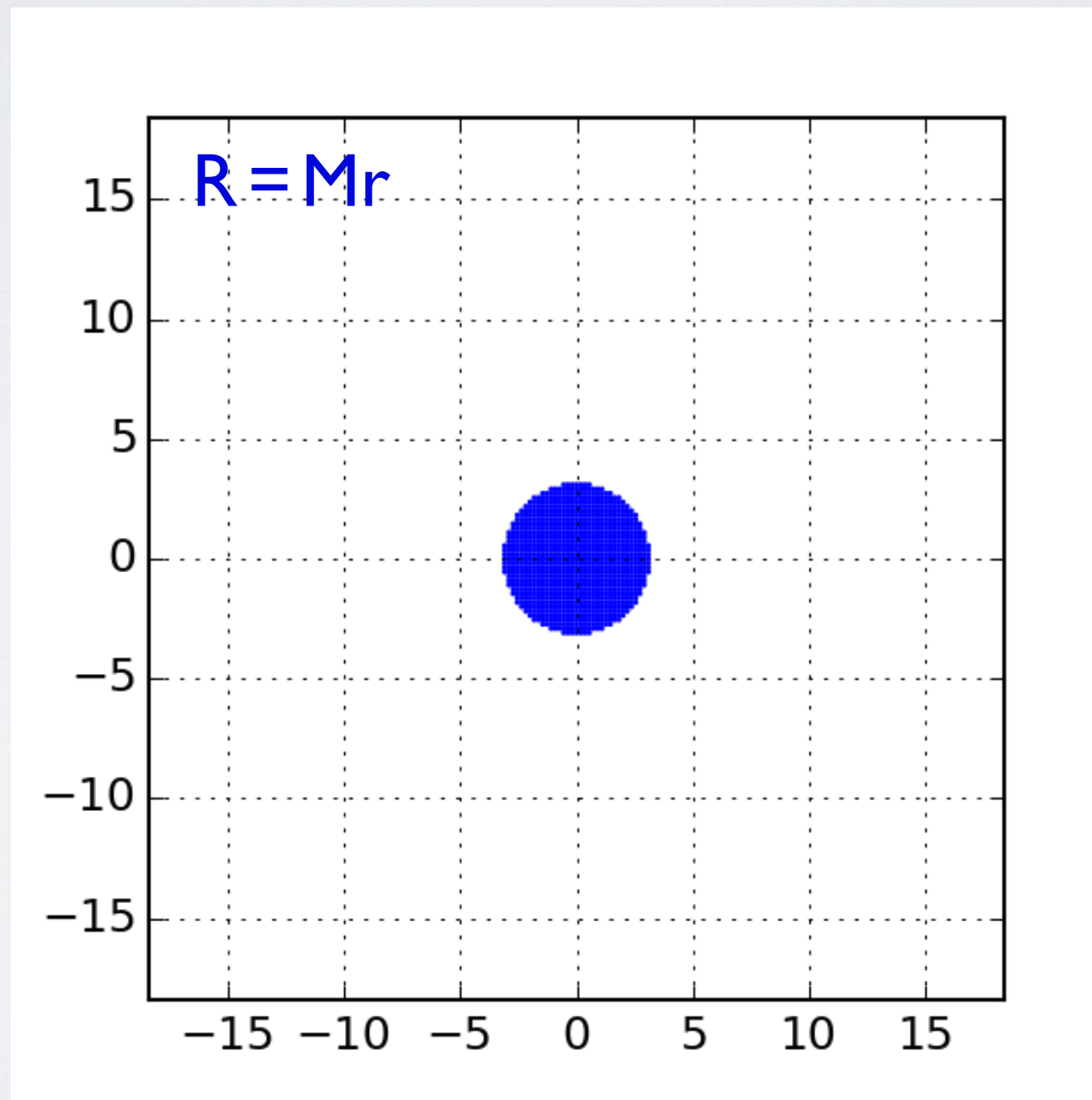
- Matrix C^{-1} **changes** the old set of coordinate axes (x, y) to a new, convenient set of coordinate axes (x', y') .



→ Figure 11.3

Meaning of Diagonalization (Ch 3.1 I)

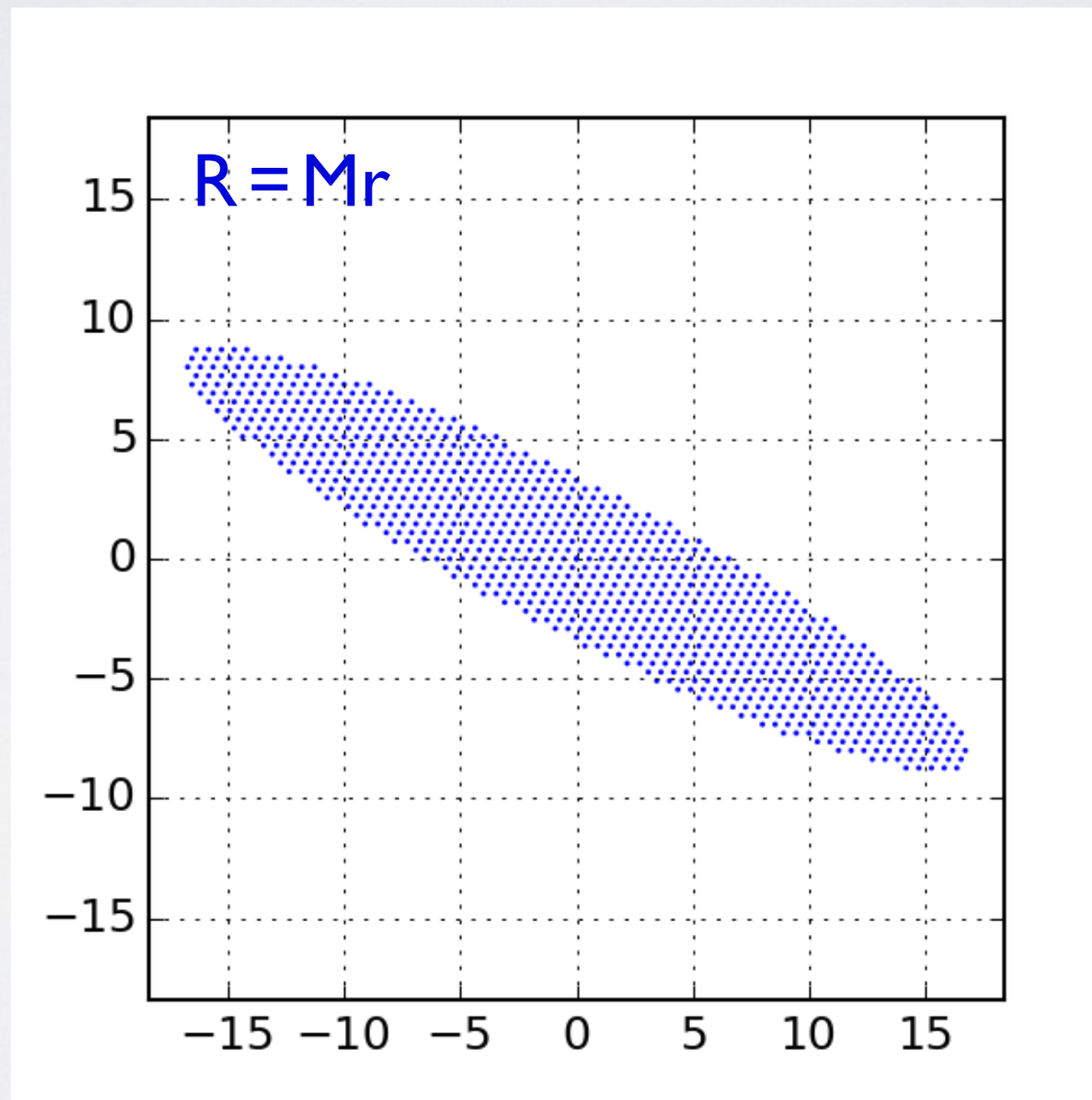
- Matrix C^{-1} **changes** the old set of coordinate axes (x, y) to a new, convenient set of coordinate axes (x', y') .



modified from dododas.github.io/linear-algebra-with-python/posts/16-12-29-2d-transformations.html

Meaning of Diagonalization (Ch 3.1 I)

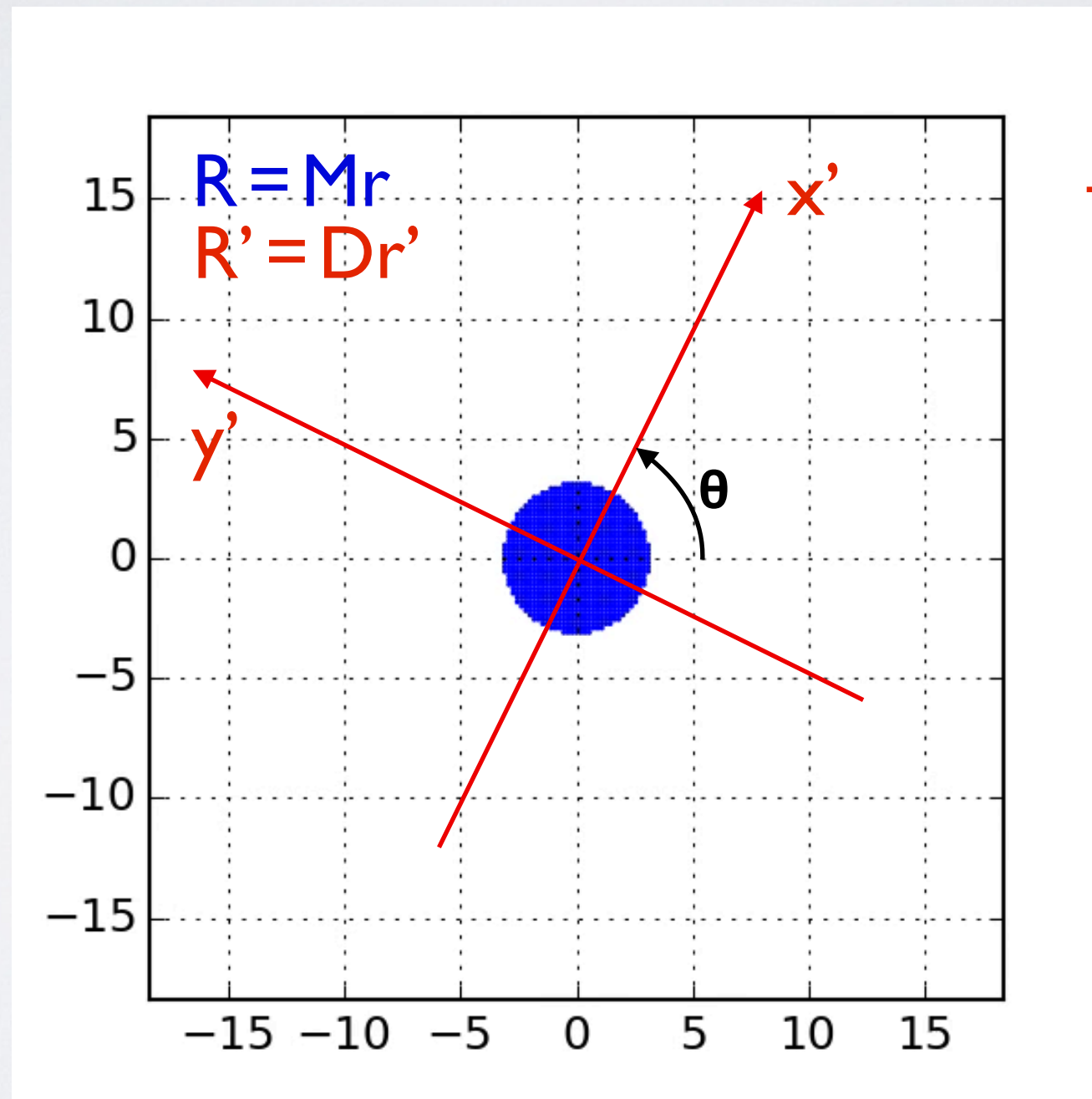
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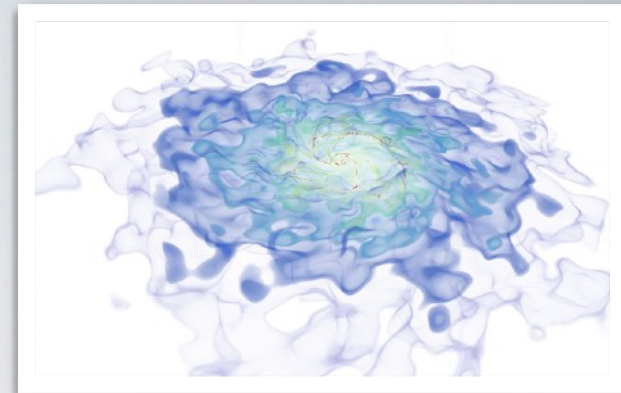
modified from dododas.github.io/linear-algebra-with-python/posts/16-12-29-2d-transformations.html

Meaning of Diagonalization (Ch 3.1 I)

- Matrix C^{-1} **changes** the old set of coordinate axes (x, y) to a new, convenient set of coordinate axes (x', y') .



→ Figure 11.3



Matrix Diagonalization On WolframAlpha

Matrix Diagonalization (Ch 3.11, Ex 1)



determinant of $\{\{1-a, -4, 2\}, \{-4, 1-a, -2\}, \{2, -2, -2-a\}\}$



Extended Keyboard

Upload

Examples

Random

Input interpretation:

$$\begin{vmatrix} 1-a & -4 & 2 \\ -4 & 1-a & -2 \\ 2 & -2 & -2-a \end{vmatrix}$$

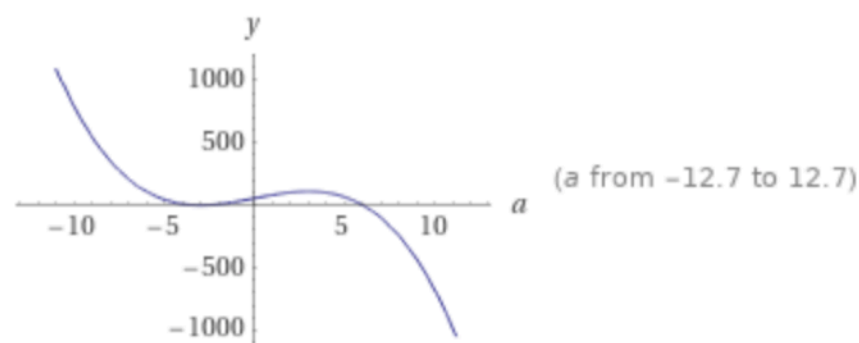
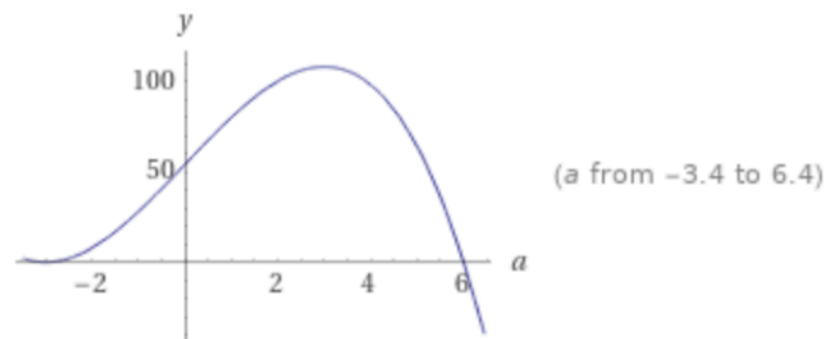
$|m|$ is the determinant

Result:

☒ Step-by-step solution

$$-a^3 + 27a + 54$$

Plots:



Matrix Diagonalization (Ch 3.11, Ex 5)



characteristic equation $(1/7)\{\{2, 6, 3\}, \{6, -3, 2\}, \{3, 2, -6\}\}$

Extended Keyboard

Upload

Examples

Random

Assuming "characteristic equation" is a math function | Use as [referring to linear algebra](#) instead

Input:

eigenvectors

$$\frac{1}{7} \begin{pmatrix} 2 & 6 & 3 \\ 6 & -3 & 2 \\ 3 & 2 & -6 \end{pmatrix}$$

Results:

[Approximate forms](#)

☒ Step-by-step solution

$$v_1 = \left(-\frac{1}{3}, 0, 1\right)$$

$$v_2 = \left(-\frac{2}{3}, 1, 0\right)$$

$$v_3 = (3, 2, 1)$$

Corresponding eigenvalues:

☒ Step-by-step solution

$$\lambda_1 = -1$$

$$\lambda_2 = -1$$

$$\lambda_3 = 1$$

Plot of eigenvectors:

Enlarge | Data | Customize | Plain Text

Matrix Diagonalization (Ch 3.1.1, Ex 3)



eigenvector {{1, sqrt(2), 1}, {-sqrt(2), 0, sqrt(2)}, {1, -sqrt(2), 1}}

Extended Keyboard

Upload

Examples

Random

Input:

eigenvectors	$\begin{pmatrix} 1 & \sqrt{2} & 1 \\ -\sqrt{2} & 0 & \sqrt{2} \\ 1 & -\sqrt{2} & 1 \end{pmatrix}$
--------------	---

Results:

[Approximate forms](#)

☒ Step-by-step solution

$$v_1 = (-1, -i\sqrt{2}, 1)$$

$$v_2 = (-1, i\sqrt{2}, 1)$$

$$v_3 = (1, 0, 1)$$

Corresponding eigenvalues:

☒ Step-by-step solution

$$\lambda_1 = 2i$$

$$\lambda_2 = -2i$$

$$\lambda_3 = 2$$

Download Page

POWERED BY THE WOLFRAM LANGUAGE

Matrix Diagonalization (Ch 10.4, Ex 5)



diagonalize{{pi,-1,-1},{-1,pi,-1},{-1,-1,pi}}

Extended Keyboard

Upload

Examples

Random

Input interpretation:

diagonalize

$$\begin{pmatrix} \pi & -1 & -1 \\ -1 & \pi & -1 \\ -1 & -1 & \pi \end{pmatrix}$$

Result:

Approximate forms

$$M = S.J.S^{-1}$$

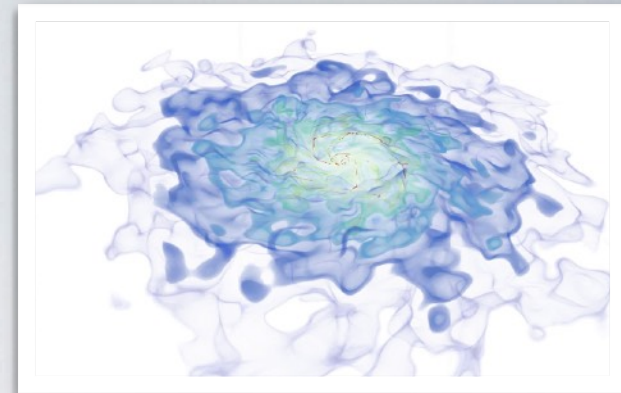
where

$$M = \begin{pmatrix} \pi & -1 & -1 \\ -1 & \pi & -1 \\ -1 & -1 & \pi \end{pmatrix}$$

$$S = \begin{pmatrix} 1 & -1 & -1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

$$J = \begin{pmatrix} \pi - 2 & 0 & 0 \\ 0 & 1 + \pi & 0 \\ 0 & 0 & 1 + \pi \end{pmatrix}$$

$$S^{-1} = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ -\frac{1}{3} & -\frac{1}{3} & \frac{2}{3} \\ -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \end{pmatrix}$$



Matrix Diagonalization On WolframAlpha: More Usages

Matrix Diagonalization (Ch 3.11, Ex 4)



inverse {{1/sqrt(2), -1/sqrt(6), 1/sqrt(3)}, {1/sqrt(2), 1/sqrt(6), -1/sqrt(3)}, {0, 2/sqrt(6), 1/sqrt(3)}}

Extended Keyboard

Upload

Examples

Random

Input:

$$\begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{pmatrix}^{-1} \quad (\text{matrix inverse})$$

Result:

[Approximate form](#)

☒ [Step-by-step solution](#)

$$\frac{1}{6} \begin{pmatrix} 3\sqrt{2} & 3\sqrt{2} & 0 \\ -\sqrt{6} & \sqrt{6} & 2\sqrt{6} \\ 2\sqrt{3} & -2\sqrt{3} & 2\sqrt{3} \end{pmatrix}$$

Expanded form:

$$\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \sqrt{\frac{2}{3}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix}$$

Dimensions:

3 (rows) × 3 (columns)

Matrix plot:

1 2 3

Enlarge | Data | Customize | Plain Text

Matrix Diagonalization (Ch 3.11, Ex 5)



det (1/7){2-7a, 6, 3}, {6, -3-7a, 2}, {3, 2, -6-7a}}



Extended Keyboard

Upload

Examples

Random

Input interpretation:

$$\left| \frac{1}{7} \begin{array}{ccc} 2-7a & 6 & 3 \\ 6 & -3-7a & 2 \\ 3 & 2 & -6-7a \end{array} \right|$$

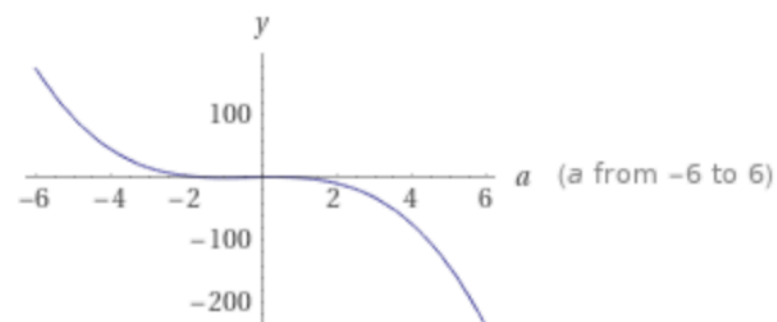
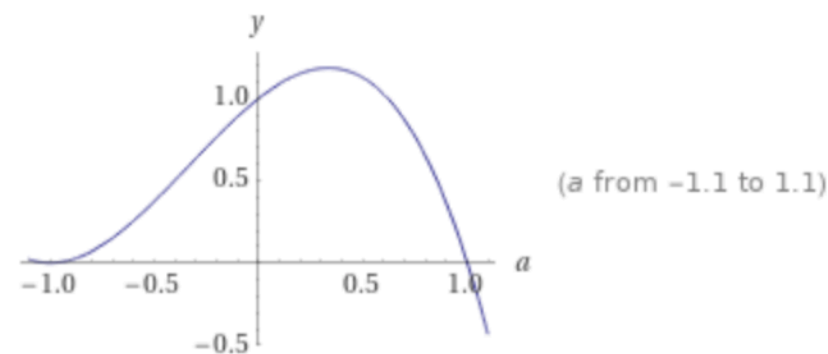
|m| is the determinant

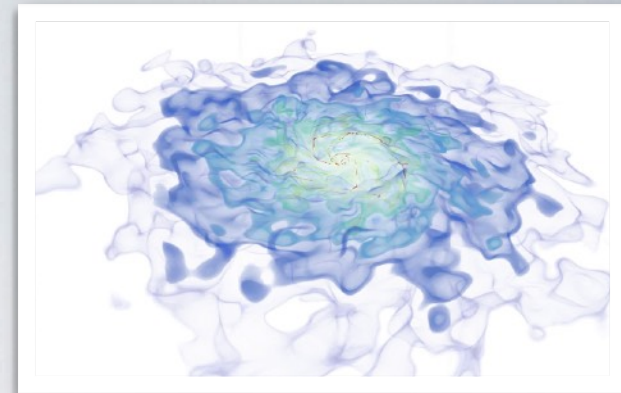
Result:

☒ Step-by-step solution

$$-a^3 - a^2 + a + 1$$

Plots:





Matrix Diagonalization: Applications

Matrix Diagonalization (Ch 3.12, Ex 1)



plot $5x^2 - 4xy + 2y^2 = 30$

NATURAL LANGUAGE

MATH INPUT

EXTENDED KEYBOARD

EXAMPLES

UPLOAD

RANDOM

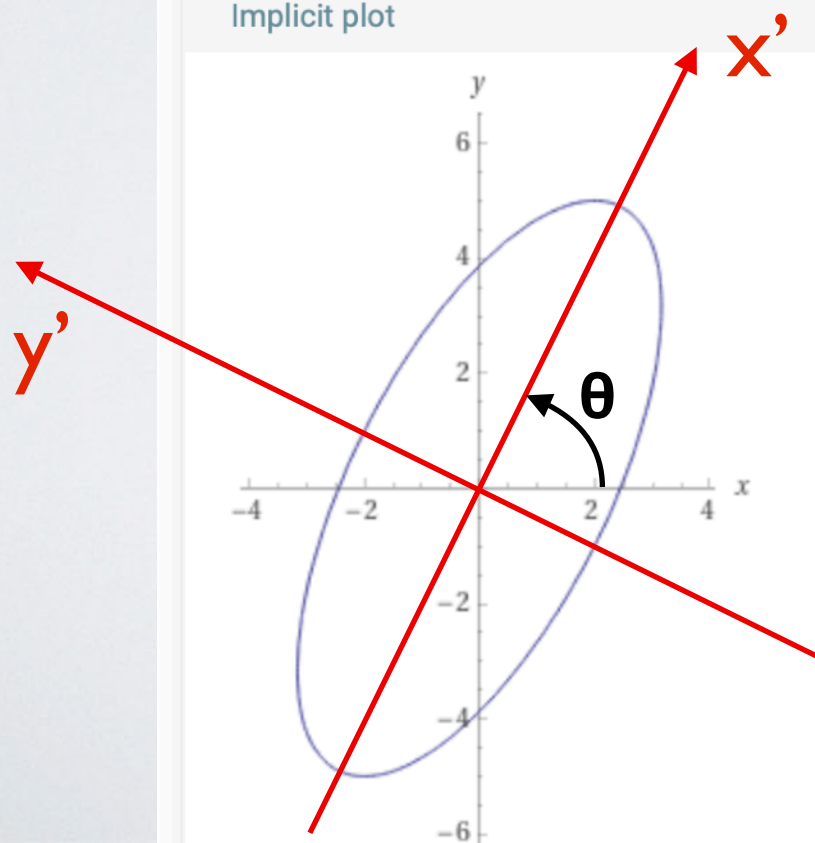
Assuming "plot" is a plotting function | Use as [referring to geometry](#) instead

Input interpretation

plot

$$5x^2 - 4xy + 2y^2 = 30$$

Implicit plot



Matrix Diagonalization (Ch 3.12, Ex 1)



plot $x^2+6xy-2y^2-2yz+z^2=24$

NATURAL LANGUAGE MATH INPUT

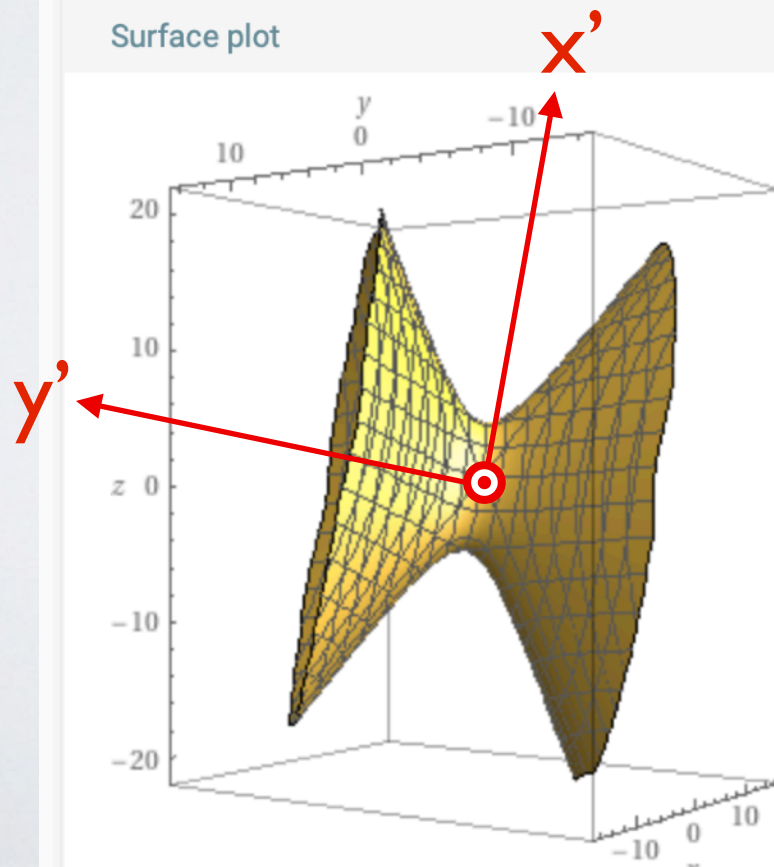
EXTENDED KEYBOARD EXAMPLES UPLOAD RANDOM

Assuming "plot" is a plotting function | Use as [referring to geometry](#) instead

Input interpretation

plot $x^2 + 6xy - 2y^2 - 2yz + z^2 = 24$

Surface plot



Matrix Diagonalization (Ch 3.12, Ex 1)



plot $x^2 + 6xy - 2y^2 - 2yz + z^2 = 24$

NATURAL LANGUAGE MATH INPUT

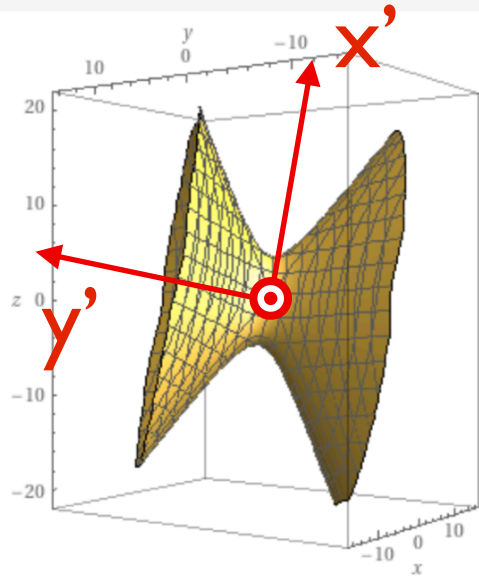
EXTENDED KEYBOARD EXAMPLES UPLOAD RANDOM

Assuming "plot" is a plotting function | Use as [referring to geometry](#) instead

Input interpretation

plot $x^2 + 6xy - 2y^2 - 2yz + z^2 = 24$

Surface plot



Geometric figure

one-sheeted hyperboloid



plot $x^2 - 4y^2 + 3z^2 = 24$

NATURAL LANGUAGE MATH INPUT

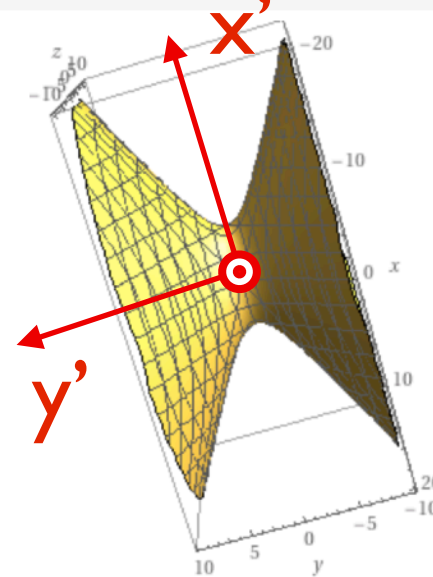
EXTENDED KEYBOARD EXAMPLES UPLOAD RANDOM

Assuming "plot" is a plotting function | Use as [referring to geometry](#) instead

Input interpretation

plot $x^2 - 4y^2 + 3z^2 = 24$

Surface plot

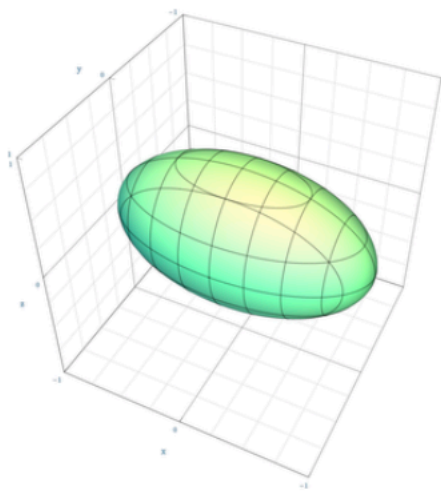


Geometric figure

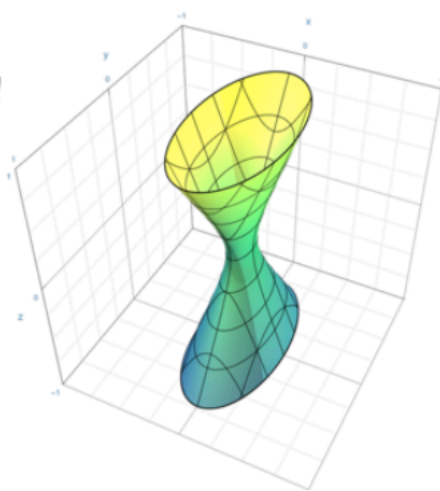
one-sheeted hyperboloid

Matrix Diagonalization (Ch 3.12, Ex 2)

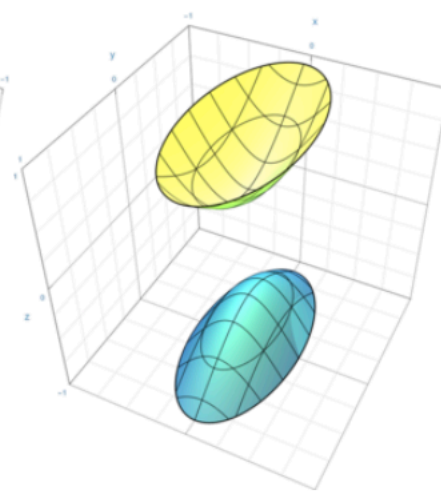
- Various quadric surfaces:



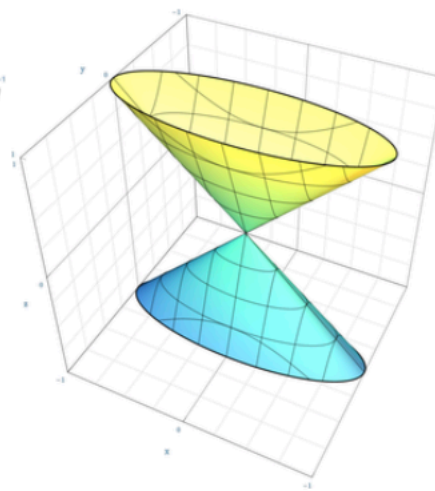
ellipsoid



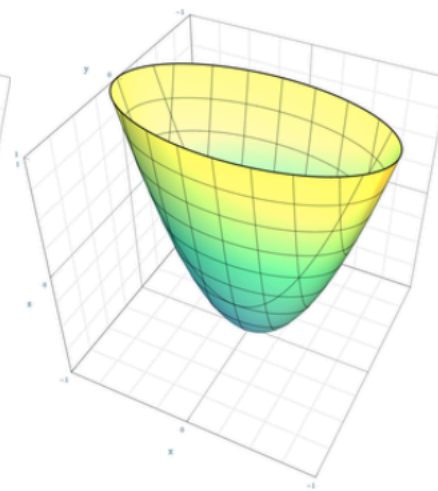
hyperboloid of
one sheet



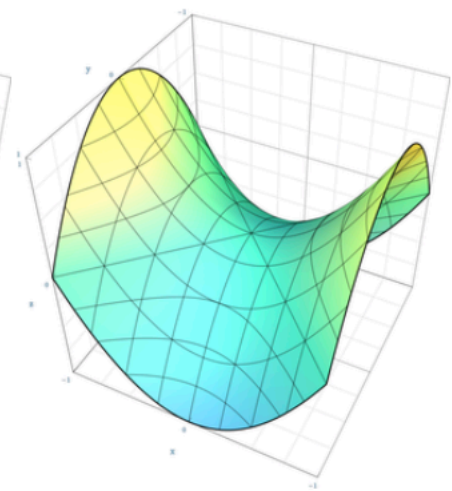
hyperboloid of
two sheets



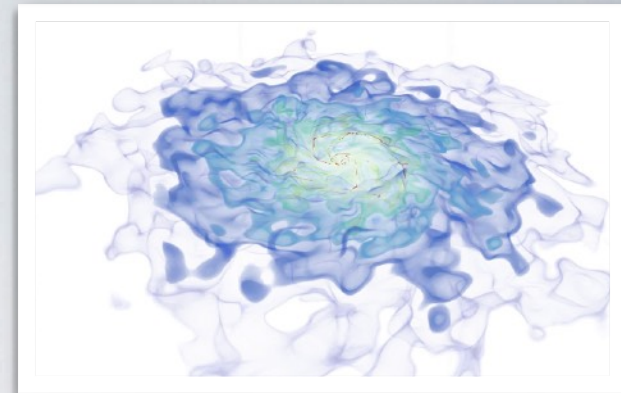
elliptic cone



elliptic
paraboloid



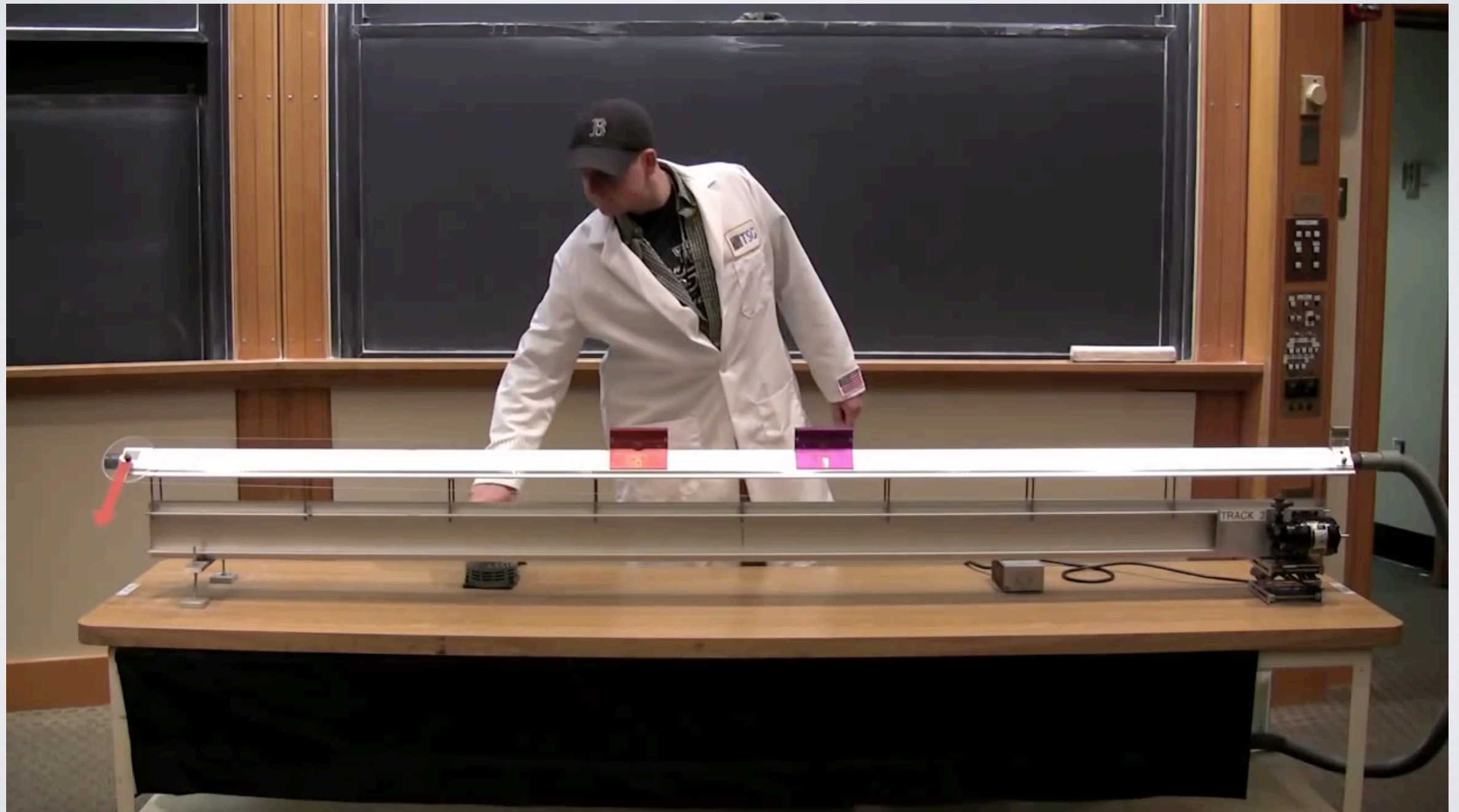
hyperbolic
paraboloid



Coupled Oscillators

Coupled Oscillators

- Two carts on a frictionless air track coupled & fixed by springs.

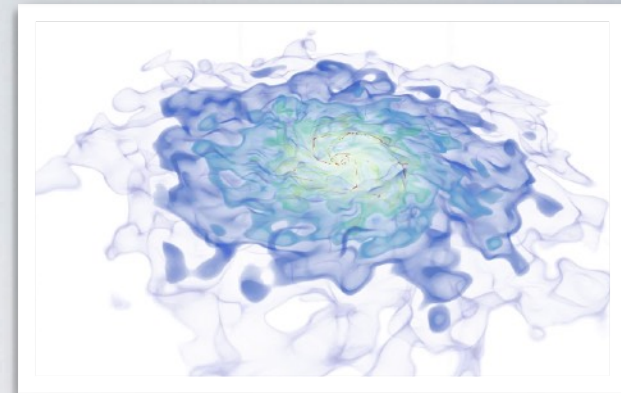


Coupled Oscillators

- Three carts on an air track coupled with four springs.



www.youtube.com/watch?v=Ye92jN6FrIU



Tensors In Physics

Tensors In Physics: Examples

- (1) Coordinate transformation tensor (Cartesian $\leftrightarrow$ Cartesian)

$$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \\ l_3 & m_3 & n_3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

Boas Ch 10.2, Eq.(2.7)

- (2) Coordinate transformation tensor (Cartesian $\leftrightarrow$ curvilinear)

$$\begin{pmatrix} dx \\ dy \\ dz \end{pmatrix} = \begin{pmatrix} \sin \theta \cos \phi & r \cos \theta \cos \phi & -r \sin \theta \sin \phi \\ \sin \theta \sin \phi & r \cos \theta \sin \phi & r \sin \theta \cos \phi \\ \cos \theta & -r \sin \theta & 0 \end{pmatrix} \begin{pmatrix} dr \\ d\theta \\ d\phi \end{pmatrix}$$

Boas Ch 10.10, Eq.(10.2)

- (3) Metric tensor

$$ds^2 = \begin{pmatrix} dx_1 & dx_2 & dx_3 \end{pmatrix} \begin{pmatrix} h_1^2 & 0 & 0 \\ 0 & h_2^2 & 0 \\ 0 & 0 & h_3^2 \end{pmatrix} \begin{pmatrix} dx_1 \\ dx_2 \\ dx_3 \end{pmatrix}$$

Boas Ch 10.2, Eq.(2.7)

Tensors In Physics: Examples

- (4) Stress tensor, Strain tensor, Elastic constant tensor

$$P_{ij} = \begin{pmatrix} P_{xx} & P_{xy} & P_{xz} \\ P_{yx} & P_{yy} & P_{yz} \\ P_{zx} & P_{zy} & P_{zz} \end{pmatrix} = C_{ijkl} S_{kl}$$

Boas Ch 10.7, Eq.(7.5)

- (5) Lorentz spacetime transformation tensor

$$\Lambda = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Griffiths Eq.(12.117)

$$\lambda = \begin{pmatrix} \gamma & i\beta\gamma & 0 & 0 \\ -i\beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Marion Eq.(14.77)

- (6) Electromagnetic field tensor

$$F^{\mu\nu} = \begin{pmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & B_z & -B_y \\ -E_y/c & -B_z & 0 & B_x \\ -E_z/c & B_y & -B_x & 0 \end{pmatrix}$$

Griffiths Eq.(12.122)

Tensors In Physics: Examples

- (7) Ricci curvature tensor, Stress-Energy tensor

$$R_{\alpha\beta} - \frac{1}{2}g_{\alpha\beta}R = 8\pi GT_{\alpha\beta}$$

Hartle Eq.(22.51)

- (8) Inertia tensor

$$I_{xx} = \sum_i m_i (y_i^2 + z_i^2)$$

$$I_{xy} = - \sum_i m_i x_i y_i$$



$$I_{xx} = \int (y^2 + z^2) dm$$

$$I_{xy} = - \int xy dm$$

Boas Ch 10.4, Eq.(4.5)

$$\{\mathbf{I}\} = \left\{ \begin{array}{ccc} \sum_{\alpha} m_{\alpha} (x_{\alpha,2}^2 + x_{\alpha,3}^2) & - \sum_{\alpha} m_{\alpha} x_{\alpha,1} x_{\alpha,2} & - \sum_{\alpha} m_{\alpha} x_{\alpha,1} x_{\alpha,3} \\ - \sum_{\alpha} m_{\alpha} x_{\alpha,2} x_{\alpha,1} & \sum_{\alpha} m_{\alpha} (x_{\alpha,1}^2 + x_{\alpha,3}^2) & - \sum_{\alpha} m_{\alpha} x_{\alpha,2} x_{\alpha,3} \\ - \sum_{\alpha} m_{\alpha} x_{\alpha,3} x_{\alpha,1} & - \sum_{\alpha} m_{\alpha} x_{\alpha,3} x_{\alpha,2} & \sum_{\alpha} m_{\alpha} (x_{\alpha,1}^2 + x_{\alpha,2}^2) \end{array} \right\}$$

Marion Eq.(11.13a)